

Given a square grid of characters in the range `ascii[a-z]`, rearrange elements of each row alphabetically, ascending. Determine if the columns are also in ascending alphabetical order, top to bottom. Return YES if they are or NO if they are not.

### Example

**grid** = ['abc', 'ade', 'efg']

The grid is illustrated below.

```
a b c
a d e
e f g
```

The rows are already in alphabetical order. The columns a a e, b d f and c e g are also in alphabetical order, so the answer would be YES. Only elements within the same row can be rearranged. They cannot be moved to a different row.

### Function Description

Complete the *gridChallenge* function in the editor below.

*gridChallenge* has the following parameter(s):

- *string grid[n]*: an array of strings

### Returns

- *string*: either YES or NO

### Input Format

The first line contains *t*, the number of testcases.

Each of the next *t* sets of lines are described as follows:

- The first line contains *n*, the number of rows and columns in the grid.
- The next *n* lines contains a string of length *n*

### Constraints

$$1 \leq t \leq 100$$

$$1 \leq n \leq 100$$

Each string consists of lowercase letters in the range `ascii[a-z]`

### Output Format

For each test case, on a separate line print YES if it is possible to rearrange the grid alphabetically ascending in both its rows and columns, or NO otherwise.

### Sample Input

| STDIN | Function                                            |
|-------|-----------------------------------------------------|
| ----- | -----                                               |
| 1     | t = 1                                               |
| 5     | n = 5                                               |
| ebacd | grid = ['ebacd', 'fghij', 'olmkn', 'trpq', 'xywuv'] |
| fghij |                                                     |
| olmkn |                                                     |

trpqs  
xywuv

## Sample Output

YES

## Explanation

The **5x5** grid in the **1** test case can be reordered to

abcde  
fghij  
klmno  
pqrst  
vwxy

This fulfills the condition since the rows 1, 2, ..., 5 and the columns 1, 2, ..., 5 are all alphabetically sorted.